Cart lab: change driving force

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use controlled data to describe how net force affects acceleration.

SUCCESS CRITERIA - I can...

1. Change driving force while keeping total mass fixed.
2. Use data or a graph to compare accelerations.
3. Explain how resistance affects the ideal relationship.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “constant” and “net force” to explain your answer. Sentence starters are optional.

constant: not changing during the stated interval

net force: the vector sum of all forces on one object

mass: a measure of how strongly an object resists acceleration

01 Warm-up: bring back an earlier idea

A puck follows a circle at a steady **6 m/s**. A student calls it equilibrium. **Draw velocity** at two points and correct the statement.

Sentence starter: In my work, I used … to show … . I labeled it … .

Marzano check - circle: ① ② ③

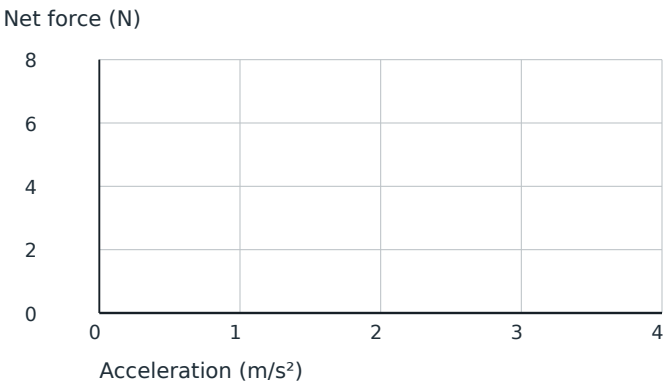
02 Worked example: explain the prediction

Reference: $a = F_{net}/M$; $F_{drive} = m_h\, g$.
Keep total moving mass M fixed. The driving weight equals net force only when resistance is negligible.

Worked reasoning: A **2 kg** cart has net forces **2** and **6 N** in two trials. Using a=F/m gives accelerations **1** and **3 m/s²**. Tripling net force triples acceleration.

Your turn: 1. **Draw a force-versus-acceleration graph** for these trials.
2. **Explain the meaning of the** slope.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: ① ② ③

03 Interpret a force graph

For total mass **1.00 kg**, use the table of ideal results to **find the slope of a** versus net force. **State its units** and recover the mass.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.

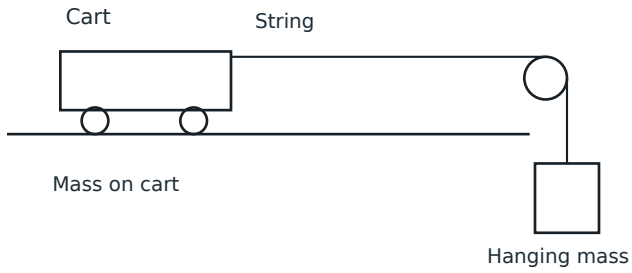
F_{net} (N)	a(m/s²)
0.245	0.245
0.490	0.490
0.735	0.735
0.980	0.980

Marzano check - circle: ① ② ③

04 Design the controlled trial

Use the setup diagram to **explain why transferring a mass** from cart to hanger changes the driving force but keeps total moving mass constant. Plan three force settings, with three acceleration trials at each setting.

Sentence starter: My design includes … . It does what is needed by … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Both variables change

The two cart trials use different setups.
1. **Use the table** to **calculate each acceleration**.
2. **Explain why tripling the net** force does not triple the acceleration in this comparison.

Sentence starter: Both … . They differ in … because … .

System	Net force (N)	Mass (kg)
A	2	1
B	6	2

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A **0.3 kg** system accelerates at **0.5 m/s²**. **Predict acceleration** for **0.7 kg** at the same net force. **Explain the effect of greater** resistance.
Use the table to keep the controlled and changed quantities distinct.

Word help: acceleration = rate of change of velocity

Sentence starter: Both … . They differ in … because … .

Trial	Mass	Net force	Acceleration
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

07 Exit check

An a-versus-F_{net} graph has slope **2.0**(m/s²)/N.
1. **Find mass**.
2. **Predict a** at **1.5 N**.
3. **Explain why plotting driving force** instead of net force can produce an offset.

Sentence starter: I predict … . I used … to make this prediction because … .

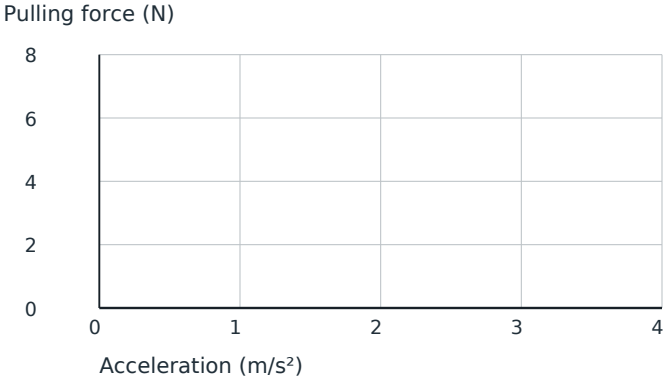
Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A **2 kg** cart is pulled with **3**, **5** and **7 N**. Its accelerations are **1**, **2** and **3 m/s²**. Plot pulling force versus acceleration. Use the slope and intercept to infer a constant opposing force.

Word help: infer = use evidence to reach a conclusion

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: ① ② ③